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Subject Name: Database Management System

Week 4.1

Module 1 : SQL Querries for practice

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Lecture – SQL querries



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Lecture – SQL querries



- We have following columns in employee table : E_id, E_name,E_salary, E_address, E_date_of_birth, E_PAN.

- **Select all columns for all employees in the table.**

```
SELECT * FROM employee;
```


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2. **Retrieve the names and salaries of all employees.**

```sql
SELECT E_name, E_salary FROM employee;
```

3. **Find the total number of employees in the table.**

```sql
SELECT COUNT(*) AS TotalEmployees FROM employee;
```

4. **List employees earning more than $50,000.**

```sql
SELECT * FROM employee WHERE E_salary > 50000;
```


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5. **Retrieve the distinct addresses of employees.**

```sql
SELECT DISTINCT E_address FROM employee;
```

6. **Calculate the average salary of all employees.**

```sql
SELECT AVG(E_salary) AS AverageSalary FROM employee;
```

7. **Find the employees who live in 'New York'.**

```sql
SELECT * FROM employee WHERE E_address = 'New York';
```


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8. **List employees born before 1990.**

```sql
SELECT * FROM employee WHERE YEAR(E_date_of_birth) < 1990;
```

9. **Find employees with a PAN card starting with 'ABC'.**

```sql
SELECT * FROM employee WHERE E_PAN LIKE 'ABC%';
```

10. **Calculate the total salary expense for the company.**

```sql
SELECT SUM(E_salary) AS TotalSalaryExpense FROM employee;
```


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11. **Retrieve the names and ages of employees.**

```sql
SELECT E_name, YEAR(NOW()) - YEAR(E_date_of_birth) AS Age FROM employee;
```

12. **List employees whose names start with 'J' and earn over $60,000.**

```sql
SELECT * FROM employee WHERE E_name LIKE 'J%' AND E_salary > 60000;
```

13. **Find employees who live in 'New York' or 'Los Angeles'.**

```sql
SELECT * FROM employee WHERE E_address IN ('New York', 'Los Angeles');
```


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14. **Calculate the highest salary in the company.**

```sql
SELECT MAX(E_salary) AS HighestSalary FROM employee;
```

15. List employees with the same address.

SELECT E_address, COUNT(*) AS NumEmployees FROM employee GROUP BY E_address HAVING COUNT(*) > 1;

16. **Find employees with birthdays in the current month.**

```sql
SELECT * FROM employee WHERE MONTH(E_date_of_birth) = MONTH(NOW());
```


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17. Retrieve employees with a PAN card in common with another employee.

```
SELECT * FROM employee WHERE E_PAN IN
(SELECT E_PAN FROM employee GROUP BY E_PAN HAVING COUNT(*) > 1);
```

18. Calculate the average salary of employees living in 'Chicago'.

```
SELECT AVG(E_salary) AS AvgSalaryInChicago FROM employee WHERE
E_address = 'Chicago';
```

19. **List employees with the highest salary.**

sql Copy code

```sql
SELECT * FROM employee WHERE E_salary = (SELECT MAX(E_salary) FROM employee);
```



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20. **Find the names and ages of employees born before 1985 and earning over $50,000.**

```sql
SELECT E_name, YEAR(NOW()) - YEAR(E_date_of_birth) AS Age
FROM employee
WHERE YEAR(E_date_of_birth) < 1985 AND E_salary > 50000;
```

21. **Retrieve the names of employees who have PAN cards and addresses starting with 'A'.**

```sql
SELECT E_name FROM employee WHERE E_PAN IS NOT NULL AND E_address LIKE 'A%';
```

22. **Calculate the total number of employees by address.**

```sql
SELECT E_address, COUNT(*) AS NumEmployees FROM employee GROUP BY E_address;
```


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23. **Find the employees with the lowest salary.**

```sql
SELECT * FROM employee WHERE E_salary = (SELECT MIN(E_salary) FROM employee);
```

24. **List employees born on the same day.**

```sql
SELECT E_date_of_birth, COUNT(*) AS NumEmployees
FROM employee
GROUP BY E_date_of_birth
HAVING COUNT(*) > 1;
```


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25. **Find employees who don't have PAN cards.**

```sql
SELECT * FROM employee WHERE E_PAN IS NULL;
```

26. Calculate the total salary expense for employees in 'New York'.

SELECT SUM(E_salary) AS TotalSalaryExpenseInNY FROM employee
WHERE E_address = 'New York';

27. **List employees whose names contain both 'John' and 'Smith'.**

```sql
SELECT * FROM employee WHERE E_name LIKE '%John%' AND E_name LIKE '%Smith%';
```


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28. **Retrieve employees with a salary greater than the average salary.**

```sql
SELECT * FROM employee WHERE E_salary > (SELECT AVG(E_salary) FROM employee);
```

29. **Find employees born before 1990 who earn more than the average salary.**

```sql
SELECT * FROM employee
WHERE YEAR(E_date_of_birth) < 1990
AND E_salary > (SELECT AVG(E_salary) FROM employee);
```


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30. **List employees who have the same address as at least one other employee and earn over $70,000.**

```
SELECT * FROM employee
WHERE E_salary > 70000  AND E_address IN
(SELECT E_address FROM employee GROUP BY E_address HAVING COUNT(*) >1);
```


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Thank You



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